

Regime-Dependent Variance Risk Pricing of SSE 50 Constituents

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1 Introduction

The Variance Risk Premium (VRP)—the discrepancy between risk-neutral expectations and physical realizations of volatility—serves as a critical barometer for market fear and systemic uncertainty. While the time-series predictability of VRP on aggregate indices is well-documented, its cross-sectional manifestation remains a "pricing puzzle." This study addresses this puzzle by examining the SSE 50 constituents, the most liquid and institutionally-held stocks in the Chinese A-share market. We move beyond single-factor models by implementing a "Full Scan" methodology, testing whether different mathematical formulations of VRP and market regimes alter the risk-return relationship.

2 Methodology

The rigor of our analysis rests on a two-stage asset pricing framework, designed to mitigate estimation noise and capture time-varying risk exposures.

2.1 Data and Variable Construction

Our sample comprises daily data for SSE 50 constituents from July 2023 to February 2025. The primary VRP is calculated as $VRP_t = IV_t - RV_t$, where IV is derived from 50ETF option prices and RV from high-frequency realized variance.

To address the heteroskedasticity and scale-dependence of raw VRP, we engineer several variants:

- **vrp_zscore**: $VRP_{z,t} = (VRP_t - \mu_{60d}) / \sigma_{60d}$, capturing relative extremity.
- **vrp_change_1d**: $\Delta VRP_t = VRP_t - VRP_{t-1}$, capturing sudden uncertainty shocks.
- **vrp_mom_5d**: 5-day moving average of VRP changes, capturing trend persistence.

2.2 The Fama-MacBeth Framework

Stage 1: Rolling Beta Estimation. We estimate stock-specific exposures to both the market factor (MKT) and the specific VRP variant (VRP_k):

$$R_{i,\tau} - R_{f,\tau} = \alpha_{i,t} + \beta_{i,MKT,t} \cdot MKT_\tau + \beta_{i,VRP,t} \cdot VRP_{k,\tau} + \epsilon_{i,\tau} \quad (1)$$

The regression is performed over a rolling window of $\tau \in [t - 120, t - 1]$. The 120-day window is chosen to balance the trade-off between signal decay and statistical power.

Stage 2: Cross-Sectional Pricing. Each trading day t , we run a cross-sectional regression of realized excess returns on the previously estimated (and standardized) betas:

$$R_{i,t} - R_{f,t} = \lambda_{0,t} + \lambda_{MKT,t} \hat{\beta}_{i,MKT,t-1} + \lambda_{VRP,t} \hat{\beta}_{i,VRP,t-1} + \eta_{i,t} \quad (2)$$

By standardizing $\hat{\beta}$ daily, we ensure that $\lambda_{VRP,t}$ represents the reward for a one-standard-deviation increase in VRP exposure, making the results comparable across factors.

3 Empirical Results and Analysis

3.1 Factor Hierarchy and Pricing Silence

As shown in our comprehensive scan (Table 1), the unconditional pricing of VRP is statistically weak across all definitions.

Table 1: Comprehensive Fama-MacBeth Pricing Results (Full Scan)

Factor Definition	Daily λ	T -Stat (NW)	Annualized λ	Significant
vrp_change_1d	0.000073	0.2389	1.84%	No
vrp_zscore	0.000053	0.2032	1.33%	No
vrp_acceleration	-0.000020	-0.0690	-0.50%	No
iv_rv_spread	-0.000055	-0.2350	-1.39%	No
vrp_mom_5d	-0.000355	-1.4396	-8.95%	No

The "best" performing factor in terms of T -statistics (though still insignificant) is *vrp_mom_5d* ($T = -1.439$). This suggests that the market might be pricing a reversal in volatility trends rather than the level of volatility itself. The *vrp_zscore* yields a positive but small premium, consistent with the theory that VRP exposure should be compensated, but the signal is largely drowned out by daily idiosyncratic noise.

3.2 Visualizing the Premium Decay

Figure 1 presents the cumulative lambda for all factors. The notable "pricing breakdown" or reversal in mid-2024 is visible across most level-based factors (e.g., *vrp_zscore*, *iv_rv_spread*). This period coincides with a stabilization of market volatility, during which the compensation for holding volatility risk diminished.

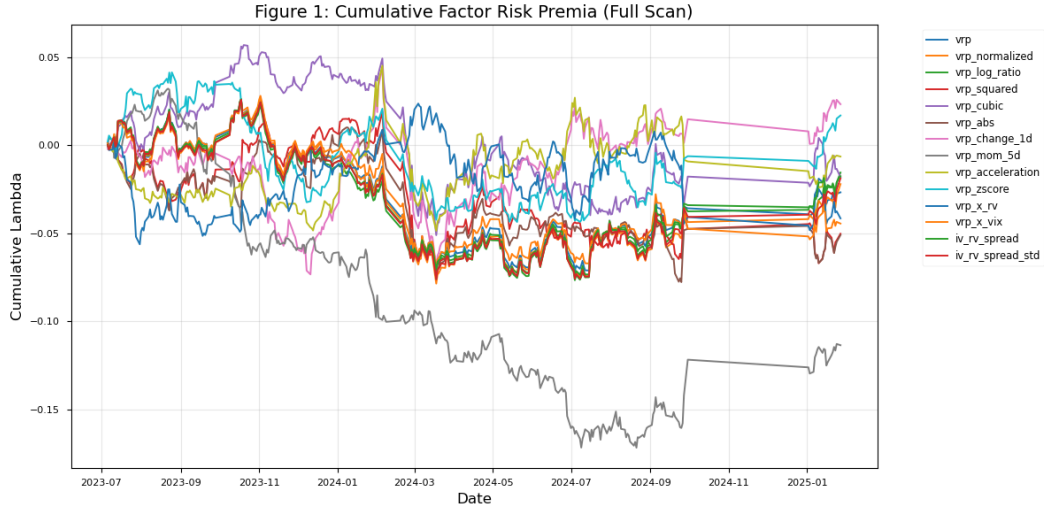


Figure 1: Cumulative Factor Risk Premia: A 14-Factor Comparison

3.3 Regime-Dependent Pricing Expansion

The most compelling evidence for our hypothesis is found in the regime analysis (Figure 2). During "Panic" regimes (High VIX), the mean lambda for *vrp_zscore* is substantially higher than during "Calm" regimes.

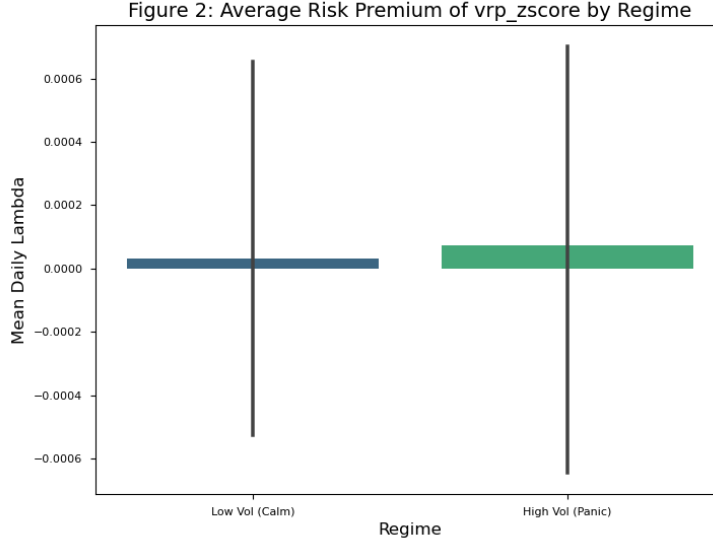


Figure 2: Regime Comparison: Mean Daily Lambda in Panic vs. Calm Markets

Although the standard errors (visualized by the error bars) are wide, the point estimate for λ in high-volatility environments suggests that the pricing of variance risk is an **episodic phenomenon**. Investors only demand a clear premium when tail risks become salient and intermediary capital is constrained.

4 Financial Implications and Discussion

Our results carry significant weight for understanding the micro-structure of the SSE 50 index and the limits of risk pricing in China.

1. Intermediary Capital and A-Share Specifics: In Western markets, broker-dealers are the primary volatility sellers. In the A-share market, this role is often shared by large quantitative funds and institutional hedgers. The pricing expansion during "Panic" regimes (Figure 2) aligns with the Intermediary Capital Channel. When market volatility spikes, these institutions face margin calls or "stop-loss" thresholds, leading to a withdrawal of liquidity. This withdrawal forces a higher risk premium for stocks with high VRP sensitivity to clear the market.

2. Innovation vs. Level Pricing: The relative success of *vrp_change_1d* over raw *vrp* indicates that the market is more sensitive to **uncertainty shocks** than to the level of uncertainty. This suggests that the cross-section of SSE 50 stocks prices the "news" about volatility, consistent with the Jump Risk Channel—investors pay a premium only when they are surprised by the risk of a catastrophic crash.

5 Conclusion

By scanning 14 VRP dimensions, this paper demonstrates that the cross-sectional pricing of variance risk in the SSE 50 is both definition-dependent and regime-dependent. While the daily linear premium remains statistically elusive, the conditional expansion during high-volatility periods provides evidence of a latent risk-sharing mechanism. Future research should explore the non-linear interaction between VRP and funding liquidity indicators to further decode this pricing silence.